

SWD392 - WEEK 6 MIDTERM CHECKPOINT

Requirements, Deliverables and Evaluation Rubric

Course	SWD392 - Software Architecture and Design	Milestone	Week 6 - MidTerm Checkpoint
Assessment	Team submission, live demo, documentation, architecture and source-code review	Impact	Contributes to project assessment and may affect Final Evaluation / final course grade

Each team must demonstrate a working and consistent system across:

Requirements -> Architecture -> Design -> Backend -> Frontend/Mobile -> Running Business Flows

1. Mandatory Architecture & Design Deliverables

Deliverable	Minimum Requirement
C4 - C0 Architecture Diagram	Overall system landscape: system boundary, users/actors, external systems and third-party services.
C4 - C1 Architecture Diagram	Major application components/containers, responsibilities, technologies and communication paths.
Use Case Diagram / User Stories	Major actors, roles, functions and business requirements.
Sequence Diagrams	Major business flows such as Order, Booking, Payment, or equivalent.
Activity Diagrams	Workflow and decision logic of major business processes.
State Diagram	Lifecycle of the main business entity, including states and transitions.

Traceability requirement: Architecture -> Business Flow -> API -> Source Code -> Data. All diagrams must reflect the current implementation.

2. Frontend Requirements

The Frontend must use a consistent UI template across major pages, provide functional navigation and role-based screens where applicable, integrate APIs for at least **5 major resources**, and support appropriate CRUD flows: **List -> Detail -> Create -> Update -> Delete**. At least one working file-upload flow must be demonstrated: **Frontend -> Backend API -> Storage**. Mock data or static UI alone is not considered complete implementation.

3. Backend Requirements

The Backend must provide **JWT Authentication, role-based Authorization enforced at API level**, more than **30 meaningful API endpoints**, business logic, validation, error handling and RESTful API conventions. Applicable list endpoints should support **Search, Sorting, Pagination and Filtering**. Endpoints created only to increase the endpoint count will not be accepted.

4. Mobile Requirements

The Mobile application must integrate with at least **5 resources** and approximately **20-30 Backend API endpoints**, use actual Backend APIs rather than hard-coded/mock data, implement at least one meaningful **Push Notification** scenario, and demonstrate three major flows: **(1) Dashboard/Overview, (2) Main Business Flow, and (3) Resource Flow** (List -> Detail -> Create/Update -> Delete/Cancel where applicable).

5. MidTerm Evaluation Rubric

Category	Weight	Evaluation Focus
C4 Architecture - C0/C1	15%	System boundary, actors, external systems, major components, technologies, interactions and implementation consistency.
UML & Software Design	15%	Use Case/User Stories, Sequence, Activity and State Diagrams.
Backend & REST API	25%	JWT, authorization, >30 APIs, business logic, REST design, Search/Sort/Paging/Filtering.
Frontend	15%	UI template, navigation, >=5 resources, CRUD, file upload and API integration.
Mobile Application	20%	>=5 resources, 20-30 APIs, Push Notification, dashboard and major business flows.
Integration & Technical Defense	10%	End-to-end operation, architecture/code traceability and technical understanding.
TOTAL	100%	

6. Checkpoint Demonstration

During the checkpoint, the lecturer may randomly request the team to explain C0/C1 architecture, locate architecture elements in source code, execute an end-to-end business flow, demonstrate authentication/authorization with different roles, execute APIs through Swagger/Postman, demonstrate CRUD and Search/Sort/Paging/Filtering, upload/retrieve a file, demonstrate Mobile API integration and Push Notification, and explain how UML diagrams correspond to implementation. **Screenshot-only evidence is not sufficient**. The lecturer may ask **any team member** to demonstrate or explain any part of the project.

7. Impact on Final Evaluation

The MidTerm Checkpoint is a **graded assessment, not a consultation session**. Results are recorded as evidence of project progress and may affect the Final Project Evaluation and final course grade. Issues identified during MidTerm are expected to be corrected before Final Evaluation. Unresolved problems, missing deliverables, or inconsistencies may result in deductions in the corresponding final rubric categories.

Before the checkpoint: Ensure the system is runnable, documentation is updated, architecture diagrams are ready, APIs can be tested, major business flows are integrated, and every team member understands the implemented solution.